

SAMIUL HOSSAIN SAJAL

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SKILLS

- Deep learning and object tracking
- Computer Vision
- Python, R and SQL
- Image segmentation of TEM datasets
- Latex, Git/Github
- Microsoft Office & Google workspace
- Spreadsheet Management
- Data Analysis and Reporting
- Research & Information Handling

LANGUAGES

- English — Fluent (C1)
- German — A1 (currently attending a German course)
- Bangla — Native

SUMMARY

Physics M.Sc. student and research assistant at Materials Data Science and Informatics (IAS-9), with a strong interest in condensed matter and materials science. Experienced in data analysis, digital documentation, AI and structured information processing.

RESEARCH & PROFESSIONAL EXPERIENCE

Student Research Assistant *(Nov 2025 - Present)*

IAS-9, Forschungszentrum Jülich

- Master's thesis focused on segmentation and tracking of in-situ TEM data.
- Grain tracking of TEM images using deep learning models.
- Collaborate with the data providor and present the progress with different groups.

Master's Thesis Research *(Oct 2019 - Nov 2021)*

Shahjalal University of Science & Technology (SUST)

- Molecular Dynamics simulation on oxidation of Aluminium nano-particles.
- Use ReaxFF force field to simulate the oxidation process in LAMMPS using HPC via linux environment.

EDUCATION

M.Sc. in Physics – *University of Cologne, Germany.* *(Oct 2022 - Present)*

Focus: Condensed Matter Physics & Computational Physics

M.Sc. in Physics – *SUST, Bangladesh.* *(Jun 2019 - Nov 2021)*

Focus: Computational Physics

B.Sc. in Physics – *SUST, Bangladesh* *(Mar 2015 - Apr 2019)*

THESIS & PRESENTATIONS

Poster Presentation: Atomic-scale investigation of the formation of Al nanoclusters in the Pulsed Laser Ablation Technique.

M.Sc. Thesis: Formation and oxidation of nano transition metals using Molecular Dynamics and ReaxFF reactive force field.

TECHNICAL TRAINING

Google Data Analytics Professional Certificate – *Coursera* *(Sep 2022)*

Python for Everybody – *Coursera* *(Feb 2025)*

Scientific Programming with Python – *University of Cologne* *(Jul 2023)*

AWARDS AND HONORS

National Science and Technology (NST) Fellowship 2020-21 *(2021)*

Honorable Photographer - "Exceed 2018-: SUST CEE Festival" *(2018)*

2nd runerup - BNMPC Science Fair 2012 *(2012)*